

DEPARTMENT OF COMPUTER SCIENCE

MCA(TYP) Semester-III

CSL5042: Advanced Computer Architecture

MID TERM EXAMINATION

564
26
38

Time Allowed : 1 Hour

Max. Marks : 30

Instruction: Attempt all Questions.

- Q1.** A computer system is designed with a memory unit of 512K words, each word containing 16 bits. Each instruction is stored in a single word and is divided into four fields: an indirect bit, an operation code, a register field that specifies one of 128 registers, and an address field.
- (a) Calculate the bits for the operation code, register, and address fields, draw the instruction format, and justify field sizes. [4 Marks]
 - (b) Calculate the number of data and address lines for the memory unit and explain their purpose. [3.5 Marks]
- Q2.** What is the instruction cycle? Draw a flowchart showing all its stages and give a brief description of each stage. [1.5, 6 Marks]
- Q3.** What are the limitations of direct cache mapping that lead to the use of set-associative mapping? Consider a 4-way set-associative cache of size 16 KB with a block size of 64 bytes and main memory of size 1 MB. Calculate the number of blocks in the cache, the number of sets, and the number of tag bits, set index bits, and block offset bits. [3, 4.5 Marks]
- Q4.** Explain pipelining with an example and describe how it improves system performance over a non-pipelined system. [7.5 Marks]